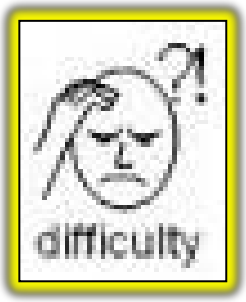
CNC Procedure 2018_03

Feb 8th 2018

[This procedure describes about how to use oscilloscope to Log the Spindle external feedback waveforms]

[Applicable for All Spindle Amplifiers]

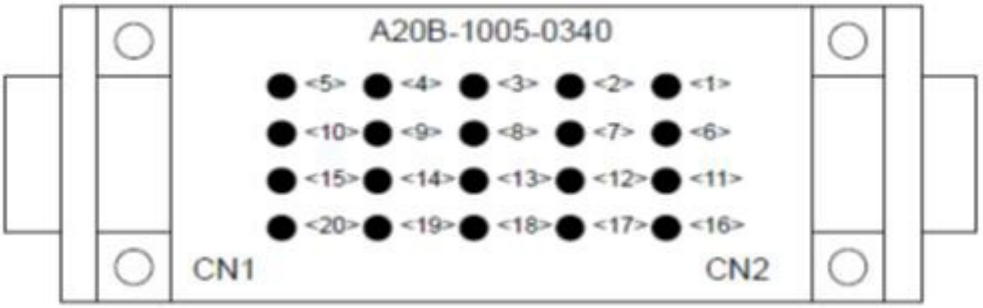
How to check Spindle Signal Online!



***How to check Spindle Feedback Sensor Signal Waveform in αi/βi* Amplifiers?**

What are the instruments used for checking?

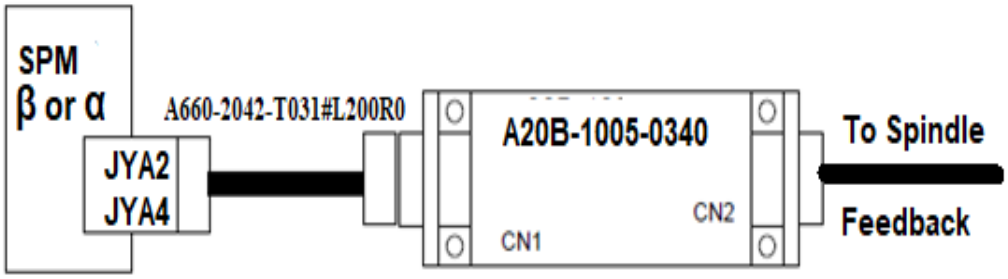
1. Oscilloscope: >2 Channel, >100MHz
2. Cable for servo check pin board: A660-2042-T031#L200R0
3. Servo check pin board BOA: A20B-1005-0340, Pin layout as below:



4. Snapshot of instruments used:

Oscilloscope	(A06B-6071-K290) Connection Cable & PCB

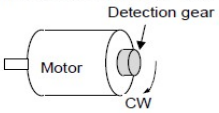
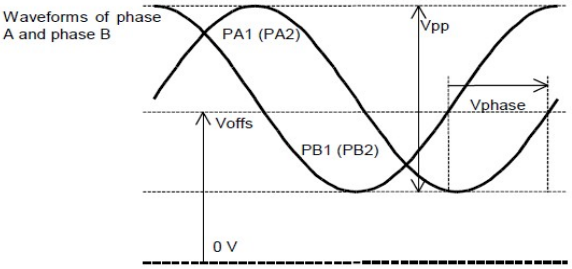
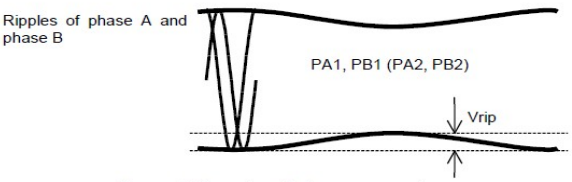
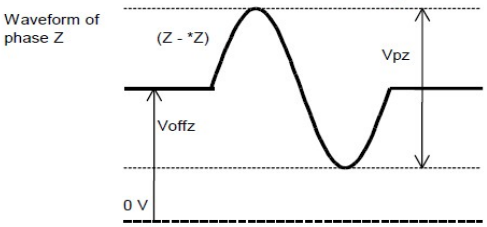
- The feedback wave form measurement connector/pins may vary depending on the configuration of the detector



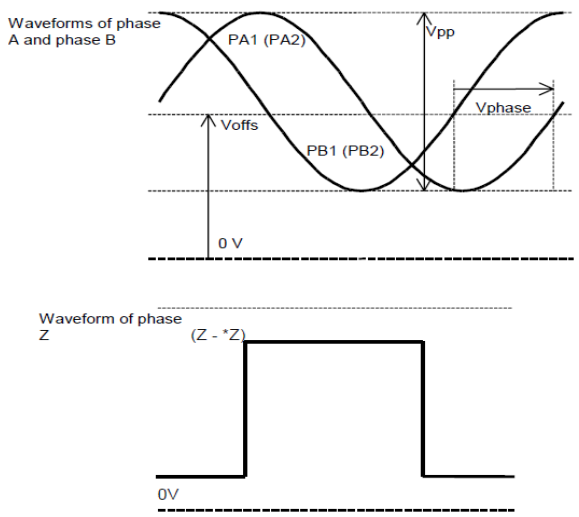
- To check the waveform with below table reference by using the check terminals pins on the check board.

Check Signal Name	Connector Name Pin Number	Sensor info
PA1 **PB1**	JYA2 (Pin5,6) JYA2 (Pin7,8)	αi M, αi MZ, and αi BZ sensors Analog αi CZ sensor
PA2 **PB2**	JYA4 (Pin5,6) JYA4 (Pin7,8)	αi BZ sensor Analog αi CZ sensor α position coder S (1024λ)
PS1	JYA2 (Pin1,2)	αi MZ and αi BZ sensors (one-rotation signal) Analog αi CZ sensor (one-rotation signal)
PS2	JYA4 (Pin1,2)	αi BZ sensor (one-rotation signal) Analog αi CZ sensor (one-rotation signal)
EXTSC1	JYA3 (Pin15)	Proximity switch (external one-rotation signal)

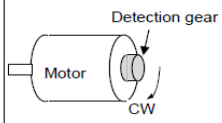
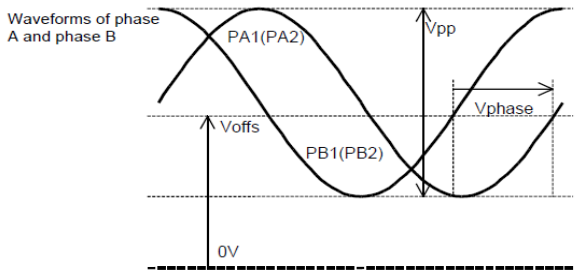
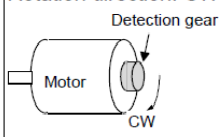
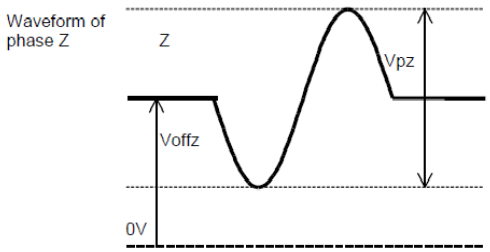
Feedback signal levels αi M sensor, αi MZ sensor, and αi BZ sensor

Measurement location	Measurement condition	Sample waveform
PA1, PB1 Separate sensors PA2, PB2	The speed must be 1500 min^{-1} or less. Rotation direction: CW 	
		
PS1 Separate sensor PS2		For αi MZ and αi BZ sensors only 

Feedback signal levels α- Position Coder

Measurement location	Measurement condition	Sample waveform
PA2, PB2	CW rotation direction as viewed from the flange	

Feedback signal levels αi-CZ Sensor

Measurement location	Measurement condition	Sample waveform
PA1, PB1 Separate sensors PA2, PB2	Speed: 500min^{-1} Rotation direction: CW 	
PS1 Separate sensor PS2	Speed: 500min^{-1} Rotation direction: CW 	

Feedback signal levels for above sample waveforms:

Measurement item	Specification	Measurement method
Vpp (for αìM/MZ/BZ)	0.5~1.2 Vp-p	Oscilloscope in DC mode
Vpp (for αìCZ)	0.9~1.2 Vp-p	
Vpp (for αPositoin coder)	0.8~1.2 Vp-p	
Voffs, Voffsz	2.5V±100mV	
Vphase	90±3º	Oscilloscope in duty mode
Vrip	<70mV	
Vpz (for αìM/MZ/BZ)	>0.5V`	
Vpz (for αìCZ)	0.66 to 1.65V p-p	

Now save all the collected data to one folder along with information of Drive & Controller name plate for future troubleshooting

FOR MORE INFORMATION ON THE ABOVE BRIEFED TOPICS PLEASE REFER RESPECTIVE AMPLIFIER MAINTENACE MANUALS